

NUMBER PATTERNS

Short Revision Notes · Grade 9 Mathematics

Remember Level — Recall Facts & Basic Knowledge

1. KEY DEFINITIONS

Term	Definition
Number Pattern	A sequence of numbers arranged in a specific order
Sequence	Another word for a number pattern
Terms	The individual numbers in a sequence
Common Difference	The constant value added (or subtracted) to get from one term to the next
General Term (T_n)	An expression that can be used to find any term in the sequence

2. COMMON DIFFERENCE FORMULA

Common difference = any term – previous term

i.e. $d = T_2 - T_1 = T_3 - T_2 = T_4 - T_3$

Sequence	Common Difference
2, 4, 6, 8, 10, ...	$4 - 2 = 2$
5, 8, 11, 14, 17, ...	$8 - 5 = 3$
10, 7, 4, 1, -2, ...	$7 - 10 = -3$
3, 3, 3, 3, 3, ...	$3 - 3 = 0$

3. GENERAL TERM (T_n) STRUCTURE

$T_n = dn + c$

where d = common difference (coefficient of n) | c = constant

Component	Meaning	Example ($T_n = 2n + 3$)
T_n	The n -th term	$2n + 3$
n	Position of the term (1st, 2nd, 3rd...)	$n = 1, 2, 3...$
Coefficient of n	The common difference	2
Constant term	The starting adjustment	3

4. FINDING TERMS USING THE GENERAL TERM

Method: Substitute the value of n into the general term.

Term Position	Calculation	Example ($T_n = 2n + 3$)
1st term (T_1)	Substitute $n = 1$	$2(1) + 3 = 5$
2nd term (T_2)	Substitute $n = 2$	$2(2) + 3 = 7$
3rd term (T_3)	Substitute $n = 3$	$2(3) + 3 = 9$
20th term (T_{20})	Substitute $n = 20$	$2(20) + 3 = 43$

5. FINDING THE TERM POSITION

Method: Set the general term equal to the given value and solve for n .

Question	Equation	Solution
Which term equals 123?	$2n + 3 = 123$	$2n = 120 \rightarrow n = 60$
Which term equals 97?	$4n - 3 = 97$	$4n = 100 \rightarrow n = 25$

6. DEVELOPING THE GENERAL TERM FROM A SEQUENCE

Step 1	Find the common difference — this becomes the coefficient of n
Step 2	Write out the multiples of the common difference
Step 3	Find what must be added or subtracted to reach the sequence
Step 4	Write the general term $T_n = dn + c$

Sequence	Common Diff.	Multiples	General Term
5, 8, 11, 14, ...	3	3, 6, 9, 12, ...	$T_n = 3n + 2$
1, 7, 13, 19, ...	6	6, 12, 18, 24, ...	$T_n = 6n - 5$
22, 19, 16, 13, ...	-3	-3, -6, -9, -12, ...	$T_n = -3n + 25$
10, 17, 24, 31, ...	7	7, 14, 21, 28, ...	$T_n = 7n + 3$

7. SPECIAL FORMULA

$$T_n = \text{first term} + (n - 1) \times \text{common difference}$$

Example: First term = 5, $d = 3$

$$\rightarrow T_n = 5 + (n-1) \times 3 = 5 + 3n - 3 = 3n + 2$$

8. KEY SYMBOLS

Symbol	Meaning
T_n	General term / n -th term
T_1	First term

Symbol	Meaning
T_2	Second term
T_3	Third term
...	The pattern continues

9. CHECK YOURSELF — QUICK QUESTIONS

No.	Question	Answer
1	What is a "term" in a number pattern?	Individual numbers in the sequence
2	What is the common difference of 4, 7, 10, 13?	3
3	What is the common difference of 8, 5, 2, -1?	-3
4	Find T_1 if $T_n = 3n + 5$	8
5	Find T_2 if $T_n = 4n - 2$	6
6	Find T_3 if $T_n = 5n + 1$	16
7	Which term is 31 if $T_n = 3n + 1$?	10th term
8	Write the general term of 2, 5, 8, 11	$T_n = 3n - 1$
9	Write the general term of 10, 7, 4, 1	$T_n = 13 - 3n$
10	The common difference is always the coefficient of what?	n

10. QUICK MEMORY AID

$$T_n = \text{First term} + (n - 1) \times \text{common difference}$$

$$T_n = dn + c \quad (d = \text{common difference}, c = \text{constant})$$

$$\text{Common difference} = T_2 - T_1 = T_3 - T_2 = T_4 - T_3$$

★ **Revision Tip:** Practice identifying common differences and writing general terms from small sequences like 5, 8, 11, 14, ... ($T_n = 3n + 2$) and 22, 19, 16, 13, ... ($T_n = 25 - 3n$).